

Erin Litzow - Research Statement

My research studies energy, environmental, and environmental health policy in low- and middle-income countries, where environmental burdens are heaviest and the evidence needed to guide policy is often thinnest. A common thread in my work is how to do credible policy analysis when the usual inputs are missing or misleading: when pollution is not monitored, when stated preferences overstate real demand, or when the benefits of a program depend on parameters no one has measured. Three principles guide my work. I aim to answer questions that directly inform current policy decisions. I collect primary data and work closely with policymakers and researchers in the countries I study. And I rely on methods that identify causal relationships while maximizing the external validity of my findings, using nationwide administrative data where it exists and multi-country designs where it does not.

Pollution, regulation, and human capital. My lead pollution research asks what happens to the communities that receive polluting industries when stricter regulation pushes them out of rich countries. In “Toxic Recycling: The Cost of Used Lead-Acid Battery Processing in Mexico” (with Zárate-Barrera, Cecato, and Romero; reject and resubmit, *Review of Economics and Statistics*), we study the January 2009 tightening of the U.S. air quality standard for lead from 1.5 to 0.15 $\mu\text{g}/\text{m}^3$, which shifted used battery recycling from the United States to Mexico. Like most developing countries, Mexico has almost no lead monitoring, so exposure cannot be observed directly. We use the locations of all 26 authorized recycling plants and every school in the country, with nationwide math and Spanish test scores from 2006 to 2013, in a spatial difference-in-differences design that compares schools within five miles of a plant to schools farther away, before and after the U.S. policy change. Math scores fell by 0.015 to 0.027 standard deviations in our preferred specification. A wind-based triple difference, which compares downwind to upwind schools near the same plant, separates lead exposure from local economic channels and yields larger effects of 0.07 standard deviations in math and 0.06 in Spanish. Effects grow over time and are concentrated in already disadvantaged communities, and foregone lifetime earnings amount to at least USD 22 million per exposed cohort. The paper is among the first to measure the human capital costs of pollution relocation caused by environmental regulation.

The costs we measure fall outside the country that made the rule, so a domestic regulatory impact analysis of the 2009 standard would typically not count them. As rich countries regulate more strictly and the energy transition expands battery and critical-mineral supply chains, policymakers in both sending and receiving countries will need tools to anticipate where these burdens go and who bears them.

This paper now anchors a research program that moves from measuring damages, to measuring exposure, to testing prevention. With a grant from the Center for Global Development, my coauthors and I are working with a health physicist, Aaron Specht of Purdue University, to measure lead directly in the bones of residents near these plants. Bone lead records cumulative exposure over years rather than a single recent dose, and it has rarely been measured at scale in low- and middle-income countries. I am also developing “Paying Not to Dump,” a field experiment in Tanzania that would test whether price incentives in the used battery collection chain can stop acid dumping and informal recycling at the source.

A related project treats climate itself as a threat to human capital. With my doctoral student, I am studying how extreme heat affects children’s learning in Kenya, Tanzania, and Uganda, combining household-based literacy and numeracy assessments with high-resolution climate reanalysis data. As warming accelerates, knowing where and for whom heat undermines learning is necessary for designing adaptation policy for education systems in the tropics.

Energy transitions and technology adoption. My second line of research studies how households in low- and middle-income countries adopt cleaner energy technologies and what policy can do to speed that transition. In “Demand for Efficient Electric Cooking Appliances in Ethiopia, Kenya, and Uganda,” funded by an Environment for Development grant on which I am co-PI, my coauthors and I elicit willingness to pay for an electric pressure cooker from 1,306 urban, grid-connected households using a

stated price ladder followed by a binding Becker-DeGroot-Marschak offer. Binding choices imply that subsidies of 82 to 91 percent of the retail price would be needed to reach 50 percent adoption. Stated willingness to pay substantially overstates revealed demand: 62 percent of households whose random price fell at or below their stated maximum refused to buy. Refusals track the purchase price rather than the appliance's two- to eight-month payback, and retail prices equal 47 to 67 percent of monthly household expenditure, a pattern consistent with liquidity constraints. A randomized information treatment on operating costs does not raise willingness to pay.

For policy design, the central finding is the gap between stated and revealed demand. Subsidy programs and cost-benefit analyses routinely rely on stated willingness to pay to project take-up and benefits. Our results show how far those projections can miss, and that information campaigns, a favored low-cost tool, do not close the gap.

Adoption is only the first step, so I also study use. In a companion pilot in Uganda, my collaborators and I installed appliance-specific smart meters in households, producing high-frequency data on actual appliance use, which self-reported surveys cannot provide. Related work includes a study of why a government biogas program in Senegal fell short of its goals and a systematic review of the impacts of modern and traditional energy use in developing countries (*Renewable and Sustainable Energy Reviews*). In a nationwide evaluation of the returns to rural electrification in Bhutan (*World Development*), we carry causal impact estimates through a full cost-benefit analysis and propagate the uncertainty in every transferred parameter. The private returns are robust to that uncertainty, while the social returns depend heavily on two inputs: the social cost of carbon and the cost of operating and maintaining the grid.

Gender and the household. A gender lens runs through my work. In the *World Bank Economic Review*, my coauthors and I follow every garment-making firm owner in a Ghanaian district capital through the COVID-19 crisis. Women whose firms closed were far more likely to exit to non-employment, while men replaced their lost income with other activities. In a multi-country impact evaluation, we estimate how improved cooking solutions change women's time use. Using a lab-in-the-field bargaining game fielded in Kenya, Malawi, Tanzania, and Zambia, we study how bargaining power within the household shapes the adoption of technologies whose benefits fall mainly to women. Policy that ignores who within the household bears the costs and receives the benefits will tend to underperform. My *Handbook of Environmental Economics* chapter on environmental health in low- and middle-income countries provides the household production framework that ties these projects together: the household is where environment and development meet.

Partnerships and next steps. My research depends on international collaboration, and I work to make those collaborations equitable. My electric cooking paper has eight authors, seven of them based in Ethiopia, Kenya, and Uganda, who led data collection in their own countries, and my Senegal study was co-led with a doctoral student at Gaston Berger University. My findings reach policymakers through the Environment for Development network, which connects environmental economics centers directly to governments in the Global South, and through the Center for Global Development's global lead-poisoning program.

Over the next several years, I will complete the bone lead study and the Toxic Recycling revision, and I plan to launch the Tanzania experiment, building toward a body of work on the economics of lead pollution in developing countries. I also plan to extend the lead research from test scores to adult income and mortality. In the energy program, I will bring the electric cooking, smart-meter, time-use, and bargaining papers to publication and develop new work on emerging cooking fuels and gender in Sub-Saharan Africa. Two newer projects extend the agenda: a meta-analysis of biogas interventions across Africa and research on urban solid waste management in African cities, which brings together the pollution and infrastructure themes of my existing work.